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Studierichting: Informatica
Oefeningenreeks 6A nr. 5.2

$$x_1 = ?? \text{ als } x_5 + x_6 = 144 \text{ en } x_8 + x_9 = 1152$$

$$\begin{cases} x_5 + x_6 = 144 \\ ar^4 + ar^5 = 144 \\ ar^4(1 + r) = 144 \end{cases} \quad \begin{cases} x_8 + x_9 = 1152 \\ ar^7 + ar^8 = 1152 \\ ar^7(1 + r) = 1152 \end{cases}$$
$$\rightarrow \frac{ar^4(1 + r)}{ar^7(1 + r)} = \frac{144}{1152}$$

$$r^3 = 8$$

$$r = 2$$

$$\Rightarrow ar^4(1 + r) = 144$$

$$a \cdot 2^4(1 + 2) = 144$$

$$48a = 144$$

$$a = 3$$

References